

Joel Valley

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EDUCATION

Toronto Metropolitan University

Bachelor of Engineering, Mechatronics (CGPA: 4.11/4.33)

Toronto, ON

Sep. 2023 – Apr. 2028

Relevant Coursework: *Microprocessor Systems, Computer Structures & Real Time Systems, Programming for Mechatronics*

Certifications: *Machine Learning Specialization (Stanford Online & DeepLearning.AI), TMU Applied Reinforcement Learning Certificate, Solidworks Certified Design Associate (CSWA)*

SKILLS

Robotics:	ROS2, Gazebo, Intel RealSense, Ubuntu Linux
Machine Learning:	PyTorch, OpenCV, Reinforcement Learning, Imitation Learning
Programming:	Python, C++, C, Matlab, Simulink, Git
CAD & Manufacturing:	Solidworks, OnShape, BambuStudio
Soft:	Leadership, Problem Solving, Communication, Team Collaboration, Technical Writing

EXPERIENCE

ELIXIR Lab

Autonomous Robotics Researcher

May 2026 – Present

- Designing an autonomous robotic manipulator system integrating perception, motion planning, and machine learning for complex task execution.
- Built a LEAP robotic hand with integrated tactile sensors for dexterous gripping tasks.
- Programmed motion planning software for a UFactory 6-DOF manipulator used for automatic data collection using ROS2.
- Used LLMs to control the arm based on predefined object locations and vague prompt inputs.
- Training custom reinforcement learning policies for arm control and grasping using NVIDIA's IsaacLab & IsaacSim.

Soft Robotics Researcher

Jan. 2025 – May 2026

- Collaborated with a team of undergraduate students on a novel untethered soft robotic crawler project for exploration.
- Optimized electromagnets for soft robotic leg actuation using OnShape and BambuStudio, reducing weight by 28%.
- Standardized design records using Excel through performance metric documentation, accelerating progress by 1 month.
- Developed a PyTorch-based regression neural network trained on synthetic data to predict robot speed from actuation parameters and achieved less than 0.0004 scaled error.

MetRocketry

Payload AI/ML Lead

Sep. 2025 – Present

- Collaborating with a 12+ member subteam to develop a computer vision deployable payload for search and rescue missions.
- Leading and mentoring team members working on AI and imaging tasks to ensure reliable model performance for the 2025-2026 competition cycle.
- Created custom aerial imagery datasets of 200+ images and trained YOLO object detection & segmentation models.
- Designed and 3D printed a custom drone camera mount using Solidworks and BambuStudio for system testing.

PROJECTS

Solar Panel PD Controller

Apr. 2026

- Using Gazebo and ROS2, created a PD controller to track the trajectory of the sun for optimal solar power output.
- Computed controller gains using natural frequency, damping ratio, and inertial parameters, and validated using simulation results.

Robotic Arm Simulation Project

Apr. 2026

- Using MATLAB implemented forward kinematics, Jacobian computation, pseudoinverse Jacobian based inverse kinematics solver, path planning, and Euler-Lagrange based dynamic simulation for a 6-DOF robotic manipulator.
- Visualized and validated correctness using MATLAB 3D plotting. Built understanding of basic robot mathematics.

Line Following & Maze Rover

Nov. 2025

- Built an Arduino-based robot capable of line following & maze navigation robot for unknown environments.
- Implemented PWM motor control and integrated LiDAR, IMU, and ultrasonic sensors for closed-loop navigation.